

WiDS 5.0 Endterm Report

Guide into the World of ML

Winter in Data Science (WiDS) 5.0

Endterm Report

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Abstract

The rapid growth of data-driven decision-making has made data science and machine learning essential skills for modern engineers and computer scientists. **Winter in Data Science (WiDS) 5.0**, organized by the Analytics Club at IIT Bombay, is an introductory program designed to bridge the gap between theoretical programming knowledge and real-world data science applications.

This report presents a detailed account of my learning journey during WiDS 5.0, covering foundational data analysis techniques using Python, exploratory data analysis on structured datasets, and the implementation of a machine learning model from scratch. The midterm phase focused on understanding datasets, performing statistical analysis, and visualizing data trends, while the endterm project involved building a linear regression model without relying on high-level machine learning libraries.

Additionally, the report discusses the role of structured learning resources such as “**Guide into the World of ML**”, challenges encountered during the program, and the key technical and conceptual skills developed. Overall, WiDS 5.0 provided a strong foundation in analytics and machine learning, motivating further exploration in this domain.

1. Introduction

Winter in Data Science (WiDS) 5.0 is a structured learning program organized by the Analytics Club with the objective of introducing students to the fundamentals of data analysis and machine learning. As a first-year undergraduate student in Computer Science and Engineering at IIT Bombay, WiDS served as my first formal exposure to applied data science workflows.

The program was designed to guide participants through a gradual learning curve, beginning with Python-based data analysis and progressing towards the implementation of a machine learning algorithm from scratch. This report documents my learning journey throughout WiDS, the projects undertaken, the challenges faced, and the skills acquired during the program.

2. Background and Motivation

Data science and machine learning have become integral components of modern technology, influencing areas such as finance, healthcare, recommendation systems, and scientific research. Despite this widespread adoption, formal exposure to applied data science is often limited during the early stages of undergraduate education.

Before joining WiDS 5.0, my experience with programming was largely theoretical and academic, primarily focused on basic syntax and problem-solving in Python. While I understood fundamental programming constructs, I lacked exposure to real-world datasets, data preprocessing techniques, and analytical workflows used in practical applications.

Furthermore, concepts such as exploratory data analysis, statistical interpretation, and machine learning were unfamiliar to me beyond surface-level definitions. I was particularly interested in understanding how mathematical models translate into working algorithms and how data-driven insights are extracted from raw datasets.

My motivation for enrolling in WiDS 5.0 was therefore multifaceted:

- To bridge the gap between theoretical programming and practical data analysis
- To understand the end-to-end workflow of a data science project
- To gain exposure to machine learning algorithms from a foundational perspective
- To develop confidence in independently exploring and analyzing datasets

WiDS provided a structured and mentor-supported environment that allowed me to gradually build these skills while reinforcing conceptual understanding through hands-on assignments.

3. Overview of WiDS Learning Structure

The WiDS program was broadly divided into two academic phases, supported by structured learning resources:

1. **Midterm Phase – Data Analysis with Python**
2. **Endterm Phase – Machine Learning Project (Linear Regression from Scratch)**

In addition to these assignments, participants were supported through a comprehensive learning roadmap titled “**Guide into the World of ML**”, which helped contextualize the overall learning journey within data science and machine learning.

4. Guide into the World of ML

4.1 Overview

“**Guide into the World of ML**” is a structured learning roadmap developed as part of the WiDS program to assist participants in navigating the field of data science and machine learning. The primary objective of this project was to provide a clear, beginner-friendly yet comprehensive guide that accommodates students with diverse backgrounds and experience levels.

The guide enabled learners to progress systematically while building strong conceptual foundations.

4.2 Learning Tracks

To ensure accessibility and effective learning, the guide was divided into **three learning tracks**:

1. **Beginner Track**
Designed for students with little or no prior experience in Python or data science. This track focused on foundational programming skills and introductory analytical thinking.
2. **Intermediate Track**
Targeted at learners with basic programming knowledge, this track emphasized data analysis using Python libraries and introduced fundamental machine learning concepts.
3. **Expert Track**
Intended for advanced learners, this track encouraged exploration of

professional-level concepts and deeper engagement with machine learning applications.

4.3 Learning Philosophy and Contribution

The guide emphasized **self-paced and flexible learning**, allowing participants to advance according to their comfort and prior experience. Content was updated regularly to maintain engagement and relevance.

The roadmap was developed by **Anubhav Dash** and **Prashant Bothra**, whose contributions ensured that the material was structured, accessible, and aligned with the objectives of the WiDS program.

4.4 Role in My WiDS Journey

This guide played a significant role in shaping my learning experience during WiDS. It provided clarity on how individual concepts fit into the broader data science pipeline and helped me transition smoothly from exploratory data analysis in the midterm phase to implementing a machine learning algorithm from scratch in the endterm project.

5. Midterm Assignment: Data Analysis with Python

5.1 Objective

The objective of the midterm assignment was to perform exploratory data analysis (EDA) using Python, derive meaningful statistical insights, and visualize patterns within the dataset.

Python-Based Data Analysis Workflow

The midterm assignment emphasized not only performing data analysis, but also following a systematic and reproducible workflow. Python was chosen as the primary programming language due to its readability, extensive ecosystem, and widespread adoption in data science.

The workflow began with importing essential libraries and setting up the development environment. Pandas was used for data manipulation due to its powerful DataFrame structure, which enables efficient handling of tabular data. NumPy supported numerical computations, while Matplotlib and Seaborn were used for visualization.

Data loading was followed by initial inspection using functions such as displaying the first few rows, checking data types, and verifying dataset dimensions. This step helped ensure that the dataset was imported correctly and that columns contained expected data formats.

Subsequently, data cleaning and validation steps were performed. Although the dataset did not contain significant missing values, checking for null entries was an important step in maintaining analytical rigor. Descriptive statistics were then computed to gain numerical insight into the dataset.

Visualization served as a bridge between numerical summaries and intuitive understanding. Each plot was carefully selected based on the type of variable being analyzed, ensuring that insights could be clearly communicated.

This structured workflow mirrors real-world data analysis practices and reinforced the importance of organization and reproducibility in analytical projects.

5.2 Dataset Used

The dataset used was a CSV file named **student_bigdata.csv**, containing both numerical and categorical attributes.

5.3 Tools and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

5.4 Tasks Performed

The following steps were undertaken:

1. Importing and inspecting the dataset
2. Analyzing dataset dimensions and structure
3. Checking for missing values
4. Computing descriptive statistics (mean, median, standard deviation)
5. Creating visualizations:
 - Histograms

- Bar charts
 - Scatter plots
6. Writing observations based on statistical and visual analysis
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5.5 Observations and Learnings

This assignment helped me understand:

- The importance of exploratory data analysis
 - How statistical summaries describe datasets
 - The role of visualizations in identifying patterns
 - The process of converting raw data into meaningful insights
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5.6 Exploratory Data Analysis Methodology

Exploratory Data Analysis (EDA) is a critical step in any data science workflow, as it enables analysts to understand the structure, quality, and underlying patterns present in the data before applying predictive models.

During the midterm assignment, EDA was performed systematically using Python and Pandas. The first step involved inspecting the dataset structure to understand column types, data ranges, and overall dimensionality. This step helped identify whether columns were numerical or categorical and informed subsequent analysis choices.

The next step involved checking for missing or null values across all columns. Handling missing data is essential, as it can significantly impact statistical summaries and visualizations. Identifying missing values early ensures that appropriate cleaning or imputation strategies can be applied if necessary.

Descriptive statistical measures such as mean, median, minimum, maximum, and standard deviation were computed for numerical columns. These measures provided a quantitative summary of the data distribution and helped identify potential outliers or skewed variables.

Visualization played a key role in EDA. Histograms were used to analyze the distribution of numerical variables, allowing insights into data spread and central tendencies. Bar charts were employed for categorical variables to understand frequency distributions, while scatter plots were used to explore relationships between pairs of numerical variables.

Through this systematic EDA process, the dataset was transformed from raw tabular data into an interpretable form, enabling informed observations and conclusions.

6. Introduction to Machine Learning

Following the midterm phase, WiDS introduced fundamental machine learning concepts, including:

- Definition of machine learning
- Supervised vs unsupervised learning
- Regression vs classification
- Importance of algorithmic understanding

Linear regression was selected for the endterm project due to its foundational importance and interpretability.

7. Endterm Project: Linear Regression from Scratch

7.1 Objective

The objective of the endterm project was to implement a **single-variable linear regression model from scratch**, without using high-level machine learning libraries.

7.2 Constraints Followed

- No use of scikit-learn
 - No use of deep learning frameworks
 - Complete manual implementation
 - Use of NumPy, Pandas, and Matplotlib only
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7.3 Dataset Used

- **Independent Variable:** SquareFeet
 - **Dependent Variable:** Price
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7.4 Mathematical Formulation of Linear Regression

Linear regression is one of the most fundamental supervised learning algorithms used for modeling the relationship between a dependent variable and one or more independent variables. In this project, a **simple linear regression model with a single predictor** was implemented.

The model is represented as:

$$y = mx + b$$

where m represents the slope of the regression line and indicates the rate of change of the dependent variable with respect to the independent variable, and b represents the intercept.

To evaluate model performance, the **Mean Squared Error (MSE)** loss function was used:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

This loss function penalizes larger errors more heavily and is widely used in regression problems.

Linear regression, despite its apparent simplicity, offers significant insight into how machine learning models learn from data. From an algorithmic perspective, the goal of linear regression is to find parameters that minimize prediction error over the entire dataset.

The slope parameter controls the sensitivity of predictions to changes in the independent variable, while the intercept represents the baseline prediction when the input is zero. Adjusting these parameters iteratively allows the model to gradually align with the underlying data distribution.

Gradient descent can be intuitively understood as a process of trial-and-error correction. Starting from arbitrary initial values, the algorithm measures how incorrect the current predictions are and adjusts parameters in small steps toward reducing this error. Repeating this process over multiple iterations leads to convergence.

Implementing this algorithm manually demystified machine learning for me. Rather than treating models as black boxes, this approach highlighted the mathematical and logical foundations underlying learning algorithms.

Gradient Descent Optimization

To minimize the loss function, gradient descent was implemented manually. Gradient descent is an iterative optimization algorithm that updates model parameters in the direction of the steepest decrease of the loss function.

The partial derivatives of the MSE with respect to m and b were derived and used to update the parameters iteratively. A suitable learning rate was selected to ensure stable convergence and prevent divergence due to large feature values.

This manual implementation provided a deeper understanding of how learning occurs in machine learning models and highlighted the importance of optimization techniques.

7.5 Results

The model successfully learned optimal values for the slope and intercept. A regression line plotted over the data points demonstrated a clear positive relationship between house size and price.

8. Discussion, Limitations, and Future Scope

While the linear regression model successfully captured the relationship between house size and price, it is important to acknowledge its limitations. The use of a single predictor variable restricts the model's ability to account for other influential factors such as location, age of property, and amenities.

Additionally, linear regression assumes a linear relationship between variables, which may not always hold true in real-world datasets. Noise, outliers, and non-linear trends can affect model performance.

Future work could involve extending the model to multiple variables, experimenting with polynomial regression, or comparing performance with other supervised learning algorithms. Incorporating feature scaling and regularization techniques could further improve model robustness.

8. Challenges Faced

Some challenges encountered during WiDS included:

- Python environment configuration
- Library dependency issues
- Understanding gradient descent intuitively
- File handling and debugging
- First-time use of Git and GitHub

Overcoming these challenges strengthened my debugging and problem-solving skills.

9. Key Takeaways

WiDS 5.0 enabled me to gain:

- Practical experience in data analysis
- A strong foundation in machine learning fundamentals
- Confidence in implementing algorithms from scratch
- Exposure to version control systems

- Improved analytical thinking
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Beyond technical knowledge, WiDS 5.0 contributed significantly to my overall skill development. The program fostered a disciplined approach to problem-solving, encouraging participants to break complex tasks into manageable components.

Working with real datasets improved my ability to reason about data quality, interpret statistical results, and communicate findings effectively. Debugging code and resolving errors strengthened my patience and analytical thinking.

The exposure to version control systems such as Git and GitHub introduced best practices used in collaborative software development. This experience will be valuable in future academic projects and internships.

Overall, WiDS helped me transition from a beginner with limited exposure to data science into a learner capable of independently exploring datasets and implementing basic machine learning models.

10. Conclusion

WiDS 5.0 has been a highly enriching learning experience. From introductory data analysis to implementing a machine learning model from first principles, the program provided a comprehensive introduction to data science.

As a first-year student at IIT Bombay, WiDS has motivated me to further explore analytics and machine learning and has laid a strong foundation for future academic and project-based work in this domain.

11. References

- WiDS 5.0 Course Material
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation

12. Acknowledgements

I would like to express my sincere gratitude to the **Analytics Club, IIT Bombay**, for organizing WiDS 5.0 and providing a structured learning platform. I also thank the mentors and contributors of the program for their guidance and continuous support throughout the journey.

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